CASE STUDY 01

Comparative Study of Popular No-Code / Low-Code Application Development Platforms

Prepared for academic learning and GitHub submission

Abstract

No-code and low-code platforms help organizations build digital applications with less hand-written code. This case study compares representative platforms across ease of use, customization, integrations, governance, and suitable use cases. The comparison is qualitative and intended to support platform selection rather than declare one platform universally best.

1. Introduction

Organizations increasingly need applications for collecting information, tracking work, managing customers, and automating routine processes. Traditional software development can require specialist skills, longer development cycles, and significant maintenance. No-code platforms use visual builders and configuration; low-code platforms add the option to write code for advanced behavior. Both can accelerate delivery, but their capabilities and limitations differ.

2. Objective and Scope

The objective is to compare common platforms and identify factors a business or student project should evaluate before selecting one. The study considers Microsoft Power Apps, Zoho Creator, Google AppSheet, Salesforce Platform, and Mendix. Features and licensing can change over time, so users should verify current plans and regional availability before making a purchasing decision.

3. Platform Overview

Microsoft Power Apps is suited to organizations already using Microsoft 365, Dataverse, and Power Automate. Zoho Creator supports form-based business applications, reports, workflows, and Deluge scripting. Google AppSheet is useful for creating mobile-friendly applications from spreadsheet-like data sources. Salesforce Platform is closely aligned with CRM and customer-service processes. Mendix targets more complex low-code development, including enterprise applications and collaboration between business and technical teams.

4. Comparative Analysis

Ease of use: AppSheet and Zoho Creator are approachable for form-and-data applications; Power Apps benefits users familiar with Microsoft tools. Customization: low-code features and scripting options allow more advanced behavior, though each platform has its own language and limits. Integrations: Power Apps works naturally with Microsoft services, Salesforce with its CRM ecosystem, and Zoho Creator with Zoho products and supported external services. Enterprise governance: identity management, permissions, auditability, deployment controls, and data policies should be examined for every platform. Portability: applications often depend on proprietary runtimes, connectors, or scripting, so moving to another platform may require rebuilding.

5. Selection Criteria

A practical evaluation should score each platform against business requirements: data sources, user count, mobile needs, offline support, authentication, role-based access, workflow complexity, reporting, integration needs, total cost, vendor support, and export options. A short proof of concept should test the most important workflow using realistic data. Include non-functional requirements such as performance, accessibility, backup, privacy, and regulatory obligations.

6. Example Scenario

A small service company needs a job-tracking app. Employees submit a service request, a supervisor assigns a technician, and managers review completion rates. Zoho Creator or AppSheet may be suitable for a quick data-driven prototype; Power Apps may be attractive if the company relies heavily on Microsoft 365; Salesforce may fit a CRM-centered process. The team should test permissions, notifications, mobile usability, and reporting before choosing.

7. Findings and Discussion

No single platform is best for every project. Ecosystem fit, integration requirements, governance, and long-term maintenance can matter more than the visual builder alone. A fast prototype is valuable, but an application used for critical operations needs documented ownership, access controls, testing, and a plan for future changes. Vendor lock-in and recurring licensing costs should be considered early.

8. Conclusion

No-code and low-code platforms make application development more accessible and can shorten delivery time. The right choice depends on the organization’s existing tools, data, required complexity, security needs, and budget. A requirements checklist, comparative proof of concept, and review of governance and total cost provide a stronger basis for selection than popularity alone.

References

Official platform documentation: Microsoft Power Apps; Zoho Creator; Google AppSheet; Salesforce Platform; Mendix. Consult each vendor’s current documentation and pricing pages for up-to-date capabilities.

Note: This document is an educational case study. Verify current platform features, syntax, pricing, and interface details in official documentation.